

FACULTAD DE INGENIERÍA DE LA UAEMEX

PROGRAM NAME

Licenciatura en Inteligencia Artificial

COURSE NAME

Experimental Design

PRACTICE NAME

Evaluation of the Effect of Mold Temperature on the Quality and
Thickness of Milk Chocolate Hemispheres Using Thermal
Monitoring with ESP32

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Objective

Evaluate the effect of silicone mold temperature on the thermal stability and visible quality of milk-chocolate half spheres through ESP32, MAX6675, K-type thermocouple monitoring, sequential and interval-based measurements, and Python variance analysis to identify the dominant induced and natural sources of variation in the molding process.

Theoretical Framework

Milk chocolate molding depends on controlled crystallization of cocoa butter while the chocolate remains fluid enough to coat the cavity before setting. In this practice, the initial temperature of the silicone mold is treated as the induced factor because it changes the heat transfer rate at the chocolate-mold interface. A colder mold removes heat faster and can thicken the wall before the chocolate spreads, while a warmer mold delays setting and can reduce wall thickness or create edge defects. Because silicone has lower thermal conductivity than metallic molds, local contact position, recovery time between cavities, and ambient laboratory conditions can produce measurable temperature variation. The experimental unit was one molded half sphere produced under one mold-temperature condition. The sampling unit was each thermocouple reading taken during the ten-second measurement window and each visual observation made on the corresponding cavity. Natural variation was expected from ambient temperature, thermocouple placement, operator timing, silicone thermal inertia, and the manual removal of excess chocolate. Induced variation was the mold thermal condition used before pouring. The Analog Devices technical datasheet specifies that the MAX6675 converts the K-type thermocouple signal to a digital temperature value with 0.25 °C resolution, and the Espressif Systems datasheet documents the ESP32 platform used to read this value through SPI before transmitting the record by USB serial [1], [2]. The dispersion of repeated readings was quantified with the sample variance, which estimates how much the measurements deviate from their mean:

$$s^2 = (1 / (n - 1)) \sum_{i=1}^n (x_i - \bar{x})^2 \quad (1)$$

In this report, equation (1) is used to compare a sequential measurement window against an interval-based window in which natural factors were allowed to change. A larger variance means that the same nominal process condition is less repeatable, even if the average temperature appears acceptable.

Equipment and Materials

Table 1 lists the equipment actually used or required to reproduce the final experiment. The revised configuration removes the previous electric-resistance heater from the final process because the chocolate was heated with a bain-marie over a gas stove, while the relay remained only as a low-voltage logic output and status element.

Element	Quantity	Relevant specification or use
ESP32 DevKit	1	USB-powered microcontroller with 3.3 V logic and programmable GPIO pins.

Element	Quantity	Relevant specification or use
MAX6675 module	1	K-type thermocouple converter with SPI-compatible digital output and 0.25 °C resolution.
K-type thermocouple	1	Metal-tip probe used to measure the silicone mold and the hot-water container.
Silicone mold for half spheres	1	Six-cavity flexible mold used as the experimental molding surface.
Milk chocolate	As required	Chocolate used for all molded half-sphere trials.
Bain-marie container and gas stove	1 set	Non-electrical heating method used to keep the water bath approximately between 39 °C and 42 °C.
24 V relay module	1	Logic output used by the monitoring circuit; it was not used to energize a high-power coffee-maker resistance.
LED indicators	3	Red, green, and blue indicators updated automatically according to the measured temperature.
LED resistors	3	330 Ω current-limiting resistors.
Decoupling capacitors	2	1 uF and 10 uF capacitors connected near the MAX6675 supply rails to reduce reading noise.
Protoboard and Dupont jumpers	1 set	Low-voltage assembly platform for the ESP32, MAX6675, LEDs, and relay signal wiring.
Computer with Arduino IDE	1	Program upload and Serial Monitor acquisition at 115200 baud.
Python with pandas, numpy, and matplotlib	1 environment	Serial capture, CSV storage, variance calculation, and graph generation.
Digital caliper or vernier	1	Recommended instrument for wall-thickness verification when the molded piece is removed.

Table 1. Equipment and materials. The table documents the final reproducible configuration and corrects the report by replacing the unused high-power resistance with a gas-heated bain-marie.

Methodology

Hardware and software configuration

The measurement circuit was assembled on a protoboard as a low-voltage system. The MAX6675 module was powered from the ESP32 supply and shared ground. The thermocouple was connected to the positive and negative terminals of the converter according to its polarity. The SPI signal lines were connected with SCK to GPIO18, CS to GPIO5, and SO to GPIO19. The red, green, and blue LEDs were connected to GPIO14, GPIO27, and GPIO25 through 330 Ω resistors. The relay input was connected as a logic signal, but the final experiment did not switch a coffee-maker resistance or any high-power load because the heating stage was performed with a bain-marie on a gas stove.

The ESP32 program read the MAX6675 temperature once per second, classified the reading into process states, updated the LED indicators, and printed a comma-separated serial line with timestamp, elapsed time, temperature, state label, and relay status. The Python capture script was designed to read those serial lines, validate that the temperature field was numeric, append each record to a CSV file, and compute mean, sample variance, and range for the selected measurement windows without copying the microcontroller code inside the report.

Chocolate molding and measurement sequence

The milk chocolate was melted and kept workable by placing the chocolate container in a hot-water bain-marie. The water container was measured with the K-type thermocouple

before and during the trials. The captured records show a hot-water interval centered near 41.6 °C, with visible readings that later dropped toward approximately 39 °C during recovery. A technical guide from Callebaut Chocolate Academy explains the importance of controlled mold handling in molded chocolates, and Callebaut's tempering guide indicates that temperature control is required to preserve the tempered structure of the chocolate [3], [4]. This range was consistent with the practical instruction to work near 39 °C to 42 °C for the available chocolate and avoided overheating the chocolate with a direct high-power resistance.

Before each molding observation, the thermocouple tip was placed on the silicone mold surface or near the selected cavity, and the serial output was recorded for approximately ten seconds. The mold photograph was used to identify cavity positions after demolding; the documented order was sample 6 at the upper-left cavity and sample 1 at the lower-right cavity. The chocolate was poured into the mold, distributed across the cavity wall, excess chocolate was removed manually, and the piece was allowed to set before visual inspection. The same operator sequence was kept as constant as possible, but contact pressure, waiting time, and local mold recovery were recorded as natural sources of variation rather than ignored.

Experimental units, sampling units, and variation sources

The experimental unit was one milk-chocolate half sphere made in one silicone cavity under a defined initial mold condition. The sampling unit was each thermocouple reading in the ten-second window and the corresponding visual quality observation for that cavity. The induced source of variation was the mold thermal condition before pouring, represented by the measured mold temperature. Natural variation came from the laboratory ambient temperature, the thermocouple contact point, the thermal inertia of silicone, the timing used to remove excess chocolate, the manual pouring technique, and the small changes in the water bath during the process.

Variation type	Source	Expected effect on the measurement or product
Induced	Initial thermal condition of the silicone mold	Changes the heat extraction rate and therefore the speed at which the chocolate wall sets.
Natural	Ambient laboratory temperature	Modifies heat loss from the mold and chocolate during pauses.
Natural	Thermocouple placement and contact pressure	Changes the local reading even if the same cavity is measured.
Natural	Silicone thermal inertia and recovery time	Produces local hot or cold zones between consecutive cavities.
Natural	Manual pouring and excess removal timing	Changes wall uniformity, rim thickness, and surface marks.
Natural	Water-bath temperature drift	Changes chocolate viscosity before pouring and can affect surface quality.

Table 2. Sources of variation. The table separates the deliberately induced factor from the natural noise sources that explain why two readings with similar averages can still produce different surfaces.

Results and Integrated Discussion

The first group of serial captures showed mold temperatures concentrated around 20.5 °C to 21.0 °C during several ten-second windows. One separate capture identified the mold as being at ambient temperature and displayed values around 23.0 °C to 23.75 °C. This means that the actual performed experiment did not cover the wide 10 °C to 25 °C treatment plan from the previous draft; the revised analysis therefore focuses on the real captured thermal range and the increase in variation when pauses and cavity changes were included.

Index	Sequential mold reading (°C)	Interval-based mold reading (°C)
1	20.50	23.50
2	20.75	23.25
3	20.75	20.50
4	21.00	20.25
5	20.50	19.75
6	20.75	20.50
7	20.50	21.25
8	20.25	20.75
9	20.75	21.00
10	20.75	20.50

Table 3. Raw thermal values transcribed from the visible serial captures. The sequential window represents consecutive readings, while the interval-based window combines readings taken after pauses or changes in cavity condition.

Variable	Sequence	Mean	Sample variance	Range
Mold temperature	Sequential	20.65 °C	0.044 °C ²	0.75 °C
Mold temperature	Interval-based	21.12 °C	1.573 °C ²	3.75 °C
Water-bath temperature	Stable hot-water segment	41.58 °C	0.112 °C ²	0.75 °C

Table 4. Mean, sample variance, and range. The mold-temperature variance increased from 0.044 °C² to 1.573 °C², approximately 35.4 times larger, when natural factors were allowed to vary.

The sequential mold window had a mean of 20.65 °C, a sample variance of 0.044 °C², and a range of 0.75 °C. This is consistent with a short measurement interval where the thermocouple remained nearly in the same position and the silicone surface had not been disturbed. The interval-based window had a mean of 21.12 °C, but its variance increased to 1.573 °C² and its range reached 3.75 °C. This increase shows that the average temperature alone is not enough to judge process stability; cavity changes, pauses, and ambient recovery can dominate the observed variation.

The hot-water segment used for the working bain-marie had a mean of 41.58 °C, a sample variance of 0.112 °C², and a range of 0.75 °C over the stable visible readings. The same screenshot also shows that the container later cooled toward the lower end of the working interval. This matters physically because chocolate viscosity and crystallization rate depend on heat exchange before pouring; a cooler bain-marie can thicken the chocolate, while a warmer one can keep it more fluid for longer.

Observed cavity position	Visible result	Interpretation
Upper-left cavity, sample 6	Incomplete or irregular rim with local edge loss	The wall likely set unevenly or was mechanically disturbed during demolding.
Upper-right cavity	Smoother surface with fewer edge residues	This cavity suggests better spreading and more uniform setting.
Middle-left cavity	Thicker and uneven rim	Manual excess removal and local cooling produced wall accumulation near the edge.
Middle-right cavity	Acceptable surface but visible rim variability	The mold temperature was close to the stable range, but pouring/removal variation remained.
Lower-left cavity	More pronounced edge defects	The final quality was sensitive to demolding and local wall thickness.
Lower-right cavity, sample 1	Visible bright patch and surface mark	The first sample likely combined temperature recovery with operator adjustment during pouring.

Table 5. Visual interpretation of the molded pieces. The table links the photograph to the physical sources of variation rather than treating the image as decoration.

The recovered temperature readings were analyzed to quantify the internal variation of each 10-second mold measurement. Sample 1 presented a mean temperature of 20.800 °C, a sample variance of 0.0667 °C², a standard deviation of 0.2582 °C, and a range of 0.75 °C. Sample 2 had the highest mean temperature, with 23.425 °C, while its variance remained moderate at 0.0424 °C². Sample 3 showed the lowest internal variation, with a variance of 0.0167 °C² and a range of only 0.25 °C, indicating the most stable thermal reading among the six samples. Sample 4 had the greatest internal dispersion, with a variance of 0.0701 °C² and a standard deviation of 0.2648 °C. Samples 5 and 6 presented intermediate behavior, with variances of 0.0618 °C² and 0.0340 °C², respectively.

The total mean temperature of the 60 recovered readings was 21.300 °C, with a total variance of 1.2623 °C², a standard deviation of 1.1235 °C, and a range of 4.00 °C. This global variance was considerably higher than the individual sample variances because it combines measurements taken at different mold temperature levels, especially Sample 2, whose mean temperature was close to 23.425 °C. Therefore, the individual sample variance is more representative of short-term measurement noise, while the total variance reflects both internal noise and between-sample thermal differences.

The molded pieces confirm that thermal repeatability is necessary but not sufficient for uniform chocolate geometry. Silicone does not distribute heat as evenly as a professional rigid mold, so the local cavity history matters. The observed rim variation and surface marks are consistent with thermal inertia, manual removal of excess chocolate, and small differences in contact temperature. The most useful process improvement would be to fix the thermocouple position, use the same waiting time before pouring, keep the water bath within a narrower window, and measure wall thickness with a caliper after demolding.

Digital Deliverables

The complete ESP32 code and the Python capture script were uploaded to a GitHub repository. The GitHub repository submitted for the project contains the program uploaded to the microcontroller, the serial reading script, a CSV file, and a README document with connection and execution instructions [5].

The YouTube demonstration video submitted for the project documents the measurement process [6].

Conclusions and Personal Learning

The objective was achieved for the thermal-variation part of the practice because the ESP32, MAX6675, and K-type thermocouple system quantified the difference between a short sequential window and an interval-based window. The evidence is the increase in mold-temperature variance from $0.044\text{ }^{\circ}\text{C}^2$ to $1.573\text{ }^{\circ}\text{C}^2$ when pauses, cavity changes, and environmental recovery were included. The dominant induced factor was the mold thermal condition, while the dominant natural factors were thermocouple placement, silicone thermal inertia, water-bath drift, and manual pouring technique.

The experiment also showed that the final quality of milk-chocolate half spheres cannot be explained only by the mean mold temperature. Several cavities showed different rim quality and surface marks despite being produced in the same general thermal range. This supports the engineering conclusion that future runs should combine thermal monitoring with standardized pouring time, fixed probe placement, and direct caliper measurements of wall thickness to connect temperature variation with the target 2 mm to 3 mm geometry.

Personally, the practice clarified why experimental design separates induced variation from natural noise. The main difficulty was that the thermocouple contact on a flexible silicone mold was not perfectly repeatable, and the team also avoided a high-power electric resistance because it was not appropriate for the chocolate process. The issue was resolved by using a gas-heated bain-marie and by interpreting the relay as a low-voltage status output rather than as the final heating actuator. This made the experiment safer and helped focus the analysis on measurement repeatability and process variation.

References

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